SHARED GRAPH STATE: PortfolioAgentState (35 Active Fields in backend/agent/state.py)

1. Investor Mandate (11)
Capital, Risk & Horizon

2. Liquid Universe (3)
suitability_passed & 504 Pool

3. Analyst Reports (4)
Macro, Market, Fund, News

4. Critic & Loop (7)
K=10..16 & rework_count<=2

5. Math & Why (10)
Whole Shares, Proofs & Trace

LLM ENGINE & DETERMINISTIC CIRCUIT-BREAKER (nodes.py)

gemini-3.8-flash
temp 0.1/0.2 • <=1.5k Tokens

Failover

fiduciary-rule-engine-v2
0.48s Zero-Latency Fallback

Stage 1: Suitability Gate (N1–N2)

N1: manager -> N2: suitability

- ✓ Capital >= ₹10,000 INR
- ✓ 2,639 -> 504 Liquid Equities
- ✓ >= 5 Stocks across >= 3 Sectors

Router D1 (route_suitability)
Pass -> N3 (Macro Analyst)
Fail -> N13 (Persistence Abort)

D1: Pass

Stage 2: Sequential Chained Specialist Analysts (N3 -> N4 -> N5 -> N6)

N3: Macro
macro_analyst_node
- Repo 6.5% • CPI 5.1%
- 10Y G-Sec 7.04%
- OUT: Cash Buffer 2.5–5%

N4: Market
market_analyst_node
- India VIX (<13/15/24)
- Nifty 50 vs 200 DMA
- OUT: Regime & Beta Cap

N5: Fundamental
fundamental_analyst
- ROE >15% • P/E [1..150]
- D/E <1.0 • Z [-3,+3]
- OUT: Quality/Val/Mom

N6: News/SEBI
news_analyst_node
- 30d News (Max 8)
- Pledges <5% & SCORES
- OUT: Sentiment [-1,+1]

Sequential State Enrichment Pipeline (Each Analyst Reads Upstream Context)
Tools: macro_data_tool -> market_technical_tool -> fundamental_research_tool -> news_filings_audit_tool

4 Reports

Stage 3: Strategist & Critic Loop (N7–N9)

N7: Lead Strategist
Selects K in [10..16] Stocks

CORE | GROWTH | DEF

K >= ceil(100 / max_stock%)

OUT: strategist_output

<=2

N8: Critic & N9: Rework
critic_node (temp=0.1)
- Audits Stock/Sector & Ethics
- N9: Swaps Flagged Tickers
- OUT: critic_approved (bool)

Router D2 (route_critic)
Approved -> N10 (SLSQP Math) | Rework (<2) -> N9 (Strategist Rework)
Exhausted (>=2) or Blocked -> N13 (Persistence Abort Commit)

D2: Approved (10–16 Stocks) -> Zero-LLM SLSQP Math (N10)

Stage 4: Zero-LLM Deterministic Mathematical Optimization & Validation (Nodes 10 & 11)

N10A: SciPy SLSQP Solver
deterministic_optimizer_node



Max Sharpe (Mod/Agg) | Min Var (Cons)
Router D3: Infeasible (<=2) -> N9 Swap

N10B: Whole-Share Allocator
allocation_rounding.py
- Q_i = floor((w_i* V_total) / P_i)
- Greedy Pass: argmax(mu_i / P_i)
- C_res = V_total - sum(Q_i P_i) >= ₹0

Computes Sharpe, Vol, VaR 95%, Max Drawdown, HHI & Div Score

N11: Post-Rounding Gate
critic_final_validation_node
- ✓ Post-Rounding Stock <= 20/25%
- ✓ Post-Rounding Sector <= 30/40%
- ✓ Unallocated Cash C_res >= ₹0

Router D4 (route_final_critic)
Pass -> N12 | Violation -> N13 Abort

D4: Pass

Stage 5: Fiduciary Explanation & ACID Persistence (N12–N13)

N12: Fiduciary Moderator
moderator_explanation_node

Level 1: Plain-English Portfolio Thesis
Macro, Regime & Risk Alignment Summary

Level 2: Per-Stock Factor Proofs
ROE, P/E, Z-Score, MCR_i & NSE/BSE Links

Tool: evidence_registry | COMPANY_THESES

PostgreSQL 16.4
ACID Commit

N13: ACID Persistence Node
persistence_node -> END

Atomic Write across 6 Tables:
portfolio_versions, security_decisions, explanations, evidence & agent_runs

ARCHITECTURAL PILLAR: Strict Separation of Qualitative AI (Nodes 3–9, 12) from Deterministic Math (Nodes 10–11)
The LLM never guesses portfolio weights or share counts: every allocation is solved by SLSQP and verified by Node 11.

End-to-End Execution Latency: ~2.14s Warm Cache | ~0.48s Deterministic Rule Fallback
All aborted/rejected paths from D1, D2, or D4 also route directly to N13 for complete audit logging.